

# Introduction to Systematic Reviews and Meta-analysis

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# Outline

Introduction

Effect sizes

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Meta-analysis: random effects model

Subgroup meta-analysis

Summary and take home message

# Introduction

# What is a systematic review?

- ▶ A systematic identification and evaluation of all the available relevant evidence
- ▶ It attempts to:
  - ▶ identify all relevant studies fitting predefined criteria
  - ▶ systematically summarize the validity and findings of the studies
  - ▶ synthesize or integrate findings
- ▶ Governed by principles of **methodological rigor**, **transparency** and **reproducibility**

# Steps in a systematic review

1. Formulate the research question
2. Decide eligibility criteria
3. Develop a protocol
4. Search for studies
5. Apply eligibility criteria to select relevant studies
6. Collect data from the studies
7. Assess studies for risk of bias/methodological quality
8. Summarize the studies
9. Synthesize findings of the studies (e.g. **meta-analysis**)
10. Interpret results and draw conclusion

# Systematic Review of RCTs

- ▶ Randomized controlled trials (RCTs) are considered the most reliable source of evidence regarding relative treatment effects.
- ▶ However, different RCTs may give different (and often conflicting) findings to the same question
- ▶ These may be due to sampling error, differences in populations, definitions of intervention, etc.

# What is a meta-analysis?

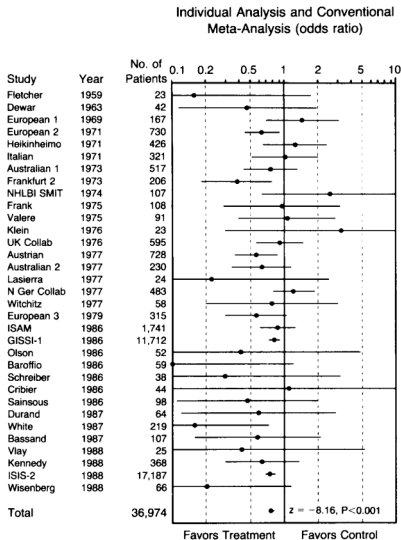
- ▶ It is a statistical method for combining the results from two or more studies with the same research question
- ▶ We can use the **PICOT** framework to identify them:
  - ▶ **P** = Population
  - ▶ **I** = Intervention
  - ▶ **C** = Control/Comparison
  - ▶ **O** = Outcome
  - ▶ **T** = Time (follow-up)
- ▶ The idea is to implement a **weighted average** according to the **precision** of each study
- ▶ It is an *optional part* of a systematic review

## Why do a meta-analysis?

- ▶ To quantify treatment effects and their uncertainty
- ▶ To increase power and precision
- ▶ To explore differences between studies

## An example

This meta-analysis included 33 RCTs performed to assess the ability of streptokinase to prevent death following a heart attack.



# Effect sizes

# Some definitions

- ▶ **Effect size:** metric to quantify the difference of an outcome between two groups
- ▶ **Treatment effect:** effect size that quantifies the impact of a medical intervention between the group of treatment and the group of control
- ▶ *How to choose the effect size?* It should be:
  1. **comparable** between studies
  2. **computable** in studies
  3. with **good technical properties**, e.g. known sample distribution
  4. **interpretable**

## Continuous outcome

### Effect size: Standardized mean difference

If the outcome is measured differently across studies, there will be multiple scales of measurement. In such cases, we need to make the effect size comparable across studies:

$$\delta = \frac{\mu_1 - \mu_2}{\sigma}$$

$$d = \frac{\bar{X}_1 - \bar{X}_2}{S}$$

$$g = J \times d$$

$$V_g = J^2 \times V_d$$

where  $J$  is a correction factor ( $0 < J < 1$ )

## Example: SMD

Let's see an example:

- ▶  $\bar{X}_1 = 103, \bar{X}_2 = 100$
- ▶  $n_1 = n_2 = 50$
- ▶  $S = 5.02$
- ▶  $J = 0.99$

Then:

$$d = \frac{\bar{X}_1 - \bar{X}_2}{S} = \frac{103 - 100}{5.02} = 0.60; V_d = 0.04$$

$$g = J \times d = 0.99 \times 0.60 = 0.59$$

$$V_g = J^2 \times V_d = (0.99)^2 \times 0.04 = 0.04$$

## Binary outcome

**Effect size: Logarithm of the ratio between two odds**

|         | Events | Non-Events | N     |
|---------|--------|------------|-------|
| Treated | $A$    | $B$        | $n_1$ |
| Control | $C$    | $D$        | $n_2$ |

where  $A = \# \text{events in treatment group}$ ,  $B = \# \text{non-events in treatment group}$ ,  $C = \# \text{events in control group}$ ,  $D = \# \text{non-events in control group}$

$$OR = \frac{\frac{A}{n_1 - A}}{\frac{C}{n_2 - C}} = \frac{\frac{A}{B}}{\frac{C}{D}} = \frac{AD}{BC}$$

$$V_{LogOR} = \frac{1}{A} + \frac{1}{B} + \frac{1}{C} + \frac{1}{D}$$

## Example: Odds Ratio

Let's see an example

|         | Dead | Alive | N   |
|---------|------|-------|-----|
| Treated | 5    | 95    | 100 |
| Control | 10   | 90    | 100 |

$$OR = \frac{AD}{BC} = \frac{5 \times 90}{95 \times 10} = 0.47$$

$$\log(OR) = \ln(0.47) = -0.76$$

$$V_{LogOR} = \frac{1}{A} + \frac{1}{B} + \frac{1}{C} + \frac{1}{D} = \frac{1}{5} + \frac{1}{95} + \frac{1}{10} + \frac{1}{90} = 0.32$$

# Meta-analysis: fixed effects model

# Meta-analysis: fixed effects model

- ▶ **Assumption:** the true effect size is the same across studies
- ▶ Observed treatment effects vary across studies due only to sampling error

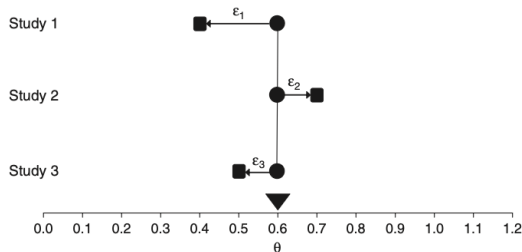


Figure: Fixed-effect model: observed vs true effect

## Fixed effects model: Effect size

Defined  $\theta$  the true effect and  $\epsilon_i$  the sampling error from study  $i$ , the treatment effect can be written as:

$$Y_i = \theta + \epsilon_i$$

The meta-analyzed estimate of the effect size comes from the following weighted mean:

$$M = \frac{\sum_{i=1}^k W_i Y_i}{\sum_{i=1}^k W_i}$$

where  $W_i = \frac{1}{V_{Y_i}}$ , with  $i = 1, \dots, k$ .  $V_{Y_i}$  is the variance of the treatment effect  $Y_i$  for study  $i$ .

## Fixed effects model: Variance and inference

The variance of the summary effect is estimated as the reciprocal of the sum of the weights:

$$V_M = \frac{1}{\sum_{i=1}^k W_i}$$

Then, 95% lower and upper limits of the confidence interval for the summary effect are:

$$LL_M = M - 1.96 \times \sqrt{V_M}$$

$$UL_M = M + 1.96 \times \sqrt{V_M}$$

## Continuous outcome: example

| Study   | Treated |    |          | Control |    |          |
|---------|---------|----|----------|---------|----|----------|
|         | Mean    | SD | <i>n</i> | Mean    | SD | <i>n</i> |
| Carroll | 94      | 22 | 60       | 92      | 20 | 60       |
| Grant   | 98      | 21 | 65       | 92      | 22 | 65       |
| Peck    | 98      | 28 | 40       | 88      | 26 | 40       |
| Donat   | 94      | 19 | 200      | 82      | 17 | 200      |
| Stewart | 98      | 21 | 50       | 88      | 22 | 45       |
| Young   | 96      | 21 | 85       | 92      | 22 | 85       |

## FE meta-analysis: an example (1)

| Study   | Effect size<br>$Y$ | Variance<br>$V_Y$ | Weight<br>$W$ | $WY$    |
|---------|--------------------|-------------------|---------------|---------|
| Carroll | 0.095              | 0.033             | 30.352        | 2.869   |
| Grant   | 0.277              | 0.031             | 32.568        | 9.033   |
| Peck    | 0.367              | 0.050             | 20.048        | 7.349   |
| Donat   | 0.664              | 0.011             | 95.111        | 63.190  |
| Stewart | 0.462              | 0.043             | 23.439        | 10.824  |
| Young   | 0.185              | 0.023             | 42.698        | 7.906   |
| Sum     |                    |                   | 244.215       | 101.171 |

## FE meta-analysis: an example (2)

$$M = \frac{101.171}{244.215} = 0.4143$$

$$V_M = \frac{1}{244.215} = 0.0041$$

$$SE_M = \sqrt{0.0041} = 0.0640$$

$$LL_M = 0.4143 - 1.96 \times 0.0640 = 0.2889$$

$$UL_M = 0.4143 + 1.96 \times 0.0640 = 0.5397$$

## FE meta-analysis: an example (3)

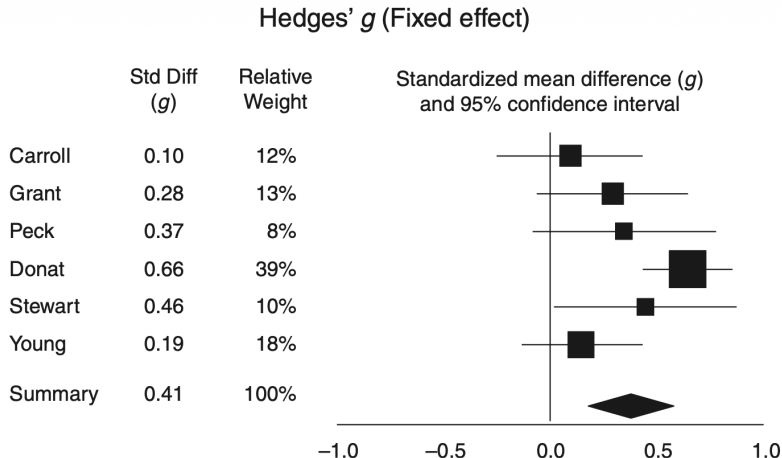


Figure: Forest plot – fixed-effect weights

# Meta-analysis: random effects model

# Within versus between study variance

- ▶ With (naturally) clustered data it is important to distinguish the variance *within* from the variance *between*
- ▶ Example of clustered data: patients from multiple studies or hospitals
- ▶ **Within-study variance** = variability inside the study, related mainly to sampling error ( $\sigma^2$ )
- ▶ **Between-study variance** = variability between studies, due to real differences between studies ( $\tau^2$ )

# Meta-analysis: random effects model

The true effect size varies from study to study, and the summary effect is the estimate of the distribution of effect sizes

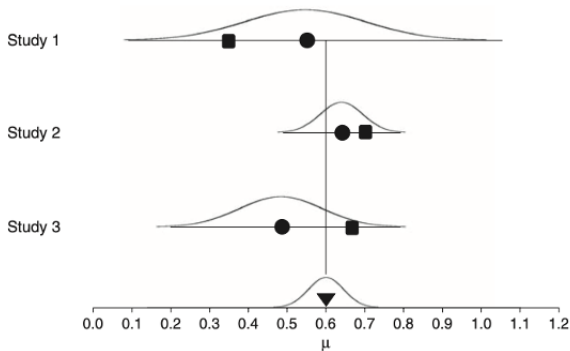


Figure: Random-effects model: between-study and within-study variance.

# Random effects model: Effect size

In each study  $i$ , the observed effect  $Y_i$  can be written as

$Y_i = \mu + \zeta_i + \epsilon_i$ , where:

- ▶  $\mu$  is the grand mean
- ▶  $\zeta_i$  is the deviation of the study's true effect (between-study variance, heterogeneity)
- ▶  $\epsilon_i$  is the deviation of the study's observed effect from the study's true effect (within-study variance)

Each study is weighted by the inverse of its variance, that includes the sampling error  $V_{Y_i}$  and the heterogeneity  $\tau^2$ :

$$V_{Y_i}^* = V_{Y_i} + \tau^2$$

$$W_i^* = \frac{1}{V_{Y_i}^*}$$

## Random effects model: Pooled estimate

The weighted mean is the sum of the effect sizes multiplied by their weights divided by the sum of weights:

$$M^* = \frac{\sum_{i=1}^k W_i^* Y_i}{\sum_{i=1}^k W_i^*}$$

The variance of the summary effect is estimated as the reciprocal of the sum of the weights:

$$V_{M^*} = \frac{1}{\sum_{i=1}^k W_i^*}$$

$$SE_{M^*} = \sqrt{V_{M^*}}$$

# Random effects model: inference and prediction

The 95% confidence interval for the summary effect is:

$$CI_{M^*} = M^* \pm 1.96 \times SE_{M^*}$$

The prediction interval is:

$$PI_{M^*} = M^* \pm t_{df}^{\alpha} \times \sqrt{T^2 + V_{M^*}}$$

where  $t_{df}^{\alpha}$  is the  $\alpha$  percentile of a t-student distribution with  $k - 2$  degrees of freedom.

The prediction interval is the interval within which a new study's effect would fall if this study was selected at random from this population

# Random effects model: heterogeneity

- ▶ There are multiple ways to estimate the heterogeneity  $\tau^2$ . The most popular are the Restricted Maximum Likelihood (REML), Paule-Mandel and DerSimonian-Laird
- ▶ How can we decide if there's heterogeneity or not? We can use the statistic  $I^2$ :

$$I^2 = \frac{\tau^2}{\tau^2 + V_Y} \times 100\%$$

- ▶  $I^2$  reflects the proportion of total variability explained by between-study variance. It ranges from 0% (all the variance is spurious) to 100% (there is heterogeneity)

## RE meta-analysis: an example (1)

Let's see an example using the same data as before. We estimated  $\tau^2$  using the methods of moments and obtained  $T^2 = 0.037$ .

| Study   | $Y$   | $V_Y$ | $V_Y + T^2$ | $W^*$  | $W^* Y$ |
|---------|-------|-------|-------------|--------|---------|
| Carroll | 0.095 | 0.033 | 0.070       | 14.233 | 1.345   |
| Grant   | 0.277 | 0.031 | 0.068       | 14.702 | 4.078   |
| Peck    | 0.367 | 0.050 | 0.087       | 11.469 | 4.204   |
| Donat   | 0.664 | 0.011 | 0.048       | 20.909 | 13.892  |
| Stewart | 0.462 | 0.043 | 0.080       | 12.504 | 5.774   |
| Young   | 0.185 | 0.023 | 0.061       | 16.466 | 3.049   |
| Sum     |       |       |             | 90.284 | 32.342  |

## RE meta-analysis: an example (2)

$$M^* = \frac{32.342}{90.284} = 0.3582$$

$$V_{M^*} = \frac{1}{90.284} = 0.0111$$

$$SE_{M^*} = \sqrt{0.0111} = 0.1052$$

$$LL_{M^*} = 0.3582 - 1.96 \times 0.1052 = 0.1520$$

$$UL_{M^*} = 0.3582 + 1.96 \times 0.1052 = 0.5645$$

$$I^2 = \frac{0.037}{0.037 +}$$

## RE meta-analysis: an example (3)

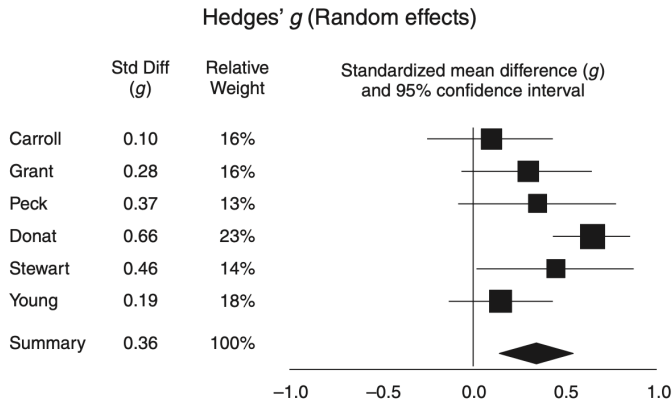


Figure: Forest plot – random-effects weights

# Fixed vs Random effect

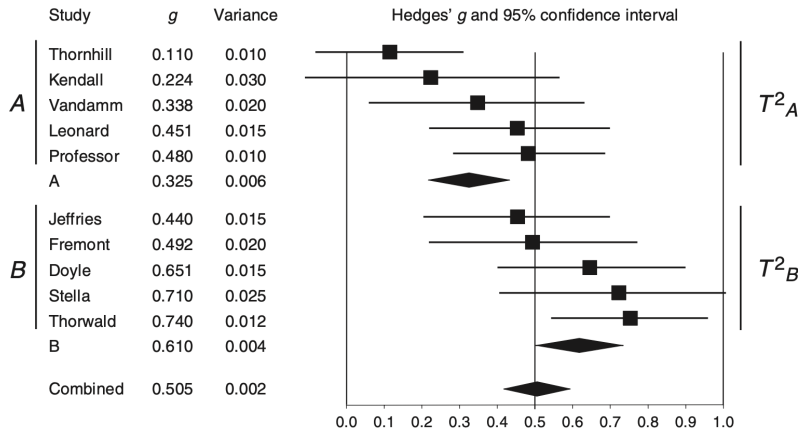
|                             | <b>Fixed effects</b>                                                                                                       | <b>Random effects</b>                                                                              |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>True effect size</b>     | Same in each study                                                                                                         | Varies across trials                                                                               |
| <b>Study contributions</b>  | Large studies contribute more, while small studies can be ignored                                                          | Small studies are not always ignored (between-study variance)                                      |
| <b>Error goes to 0 with</b> | Infinite sample size                                                                                                       | Infinite sample sizes and an infinite number of studies                                            |
| <b>To use if...</b>         | Studies are believed to be functionally identical and the goal is to evaluate the effect size for an identified population | Studies are believed to significantly differ and the goal is to generalize to a range of scenarios |

# Subgroup meta-analysis

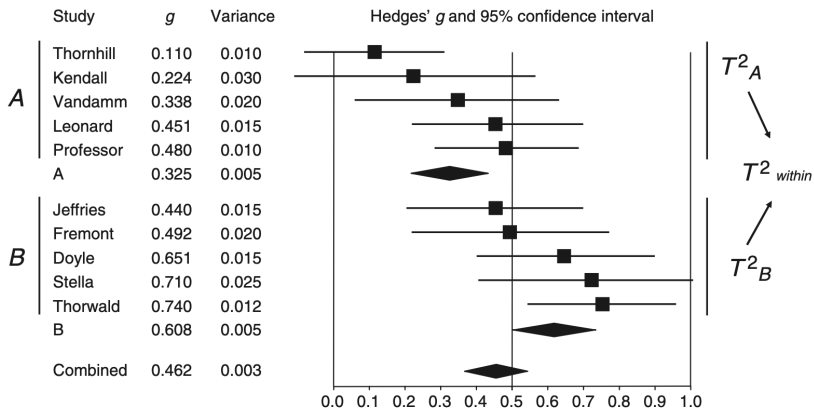
# Subgroup analysis and meta-regression

- ▶ The aim of subgroup analysis and meta-regression is to examine the association between study-level characteristics (covariates) and treatment effects
- ▶ Covariates must be decided with a scientific rationale
- ▶ The number of subgroups must be kept low to avoid issues related to multiple testing.
- ▶ Two options are available:
  1. **Fixed effect meta-analysis:** we assume that all studies in subgroup A share a common effect and studies in subgroup B share a common effect
  2. **Random effect meta-analysis:** heterogeneity  $\tau^2$  must be estimated within subgroups, then it can be pooled or not when we perform a meta-analysis.

# Subgroup analysis: random effects with separate estimates of the between-study variance



# Subgroup analysis: random effects with pooled between-study variance



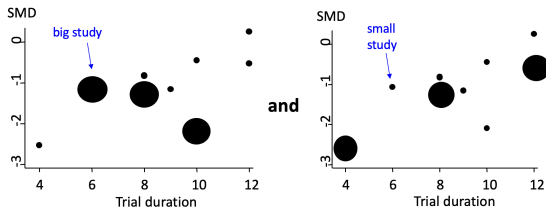
# Meta-regression

- ▶ Describes a linear relationship between **outcome variable** and **explanatory variable(s)**
- ▶ Assume study  $i$  reports  $y_i$  and has covariate  $x_i$ , a random effects meta-regression is written as:

$$y_i = \alpha + \beta x_i + \epsilon_i + \delta_i$$

where  $\epsilon_i$  is the random error and  $\delta_i$  is the random effect

- ▶ Studies have different influences on the analysis and are weighted according to their precision



# Summary and take home message

# Take home message (I)

- ▶ Start with a **clear research question**
- ▶ Compare like with like:
  - ▶ participants in one study are not directly compared with those in another
  - ▶ each study is analysed separately
  - ▶ summary statistics are combined to give the meta-analysis results

## Take home message (II)

- ▶ **Meta-analysis:** calculate a *summary* effect estimate which is a weighted average of the estimated treatment effects from individual studies
  - ▶ Weight studies according to the information they provide!
- ▶ **Fixed-effect meta-analysis:**
  - ▶ assumes treatment effect is the same in each study
  - ▶ weights:  $W_i = \frac{1}{V_{Y_i}}$
- ▶ **Random-effect meta-analysis:**
  - ▶ treatment effect varies between studies
  - ▶ weights:  $W_i^* = \frac{1}{V_{Y_i} + \tau^2}$

## To further explore

- ▶ Codes and data for the analyses in R can be retrieved in the following GitHub repository
- ▶ Suggested reading: Introduction to Meta-Analysis (Borenstein et al., 2009)